

Paul Bittner, Ph.D.

p.m.bittner@web.de



Software engineer and researcher on programming languages; longtime professional and hobby programmer; can formalize, test, and prove soundness correctness of software; values and writes high-quality documentation; enjoys engineering expressive, easy-to-use but hard-to-misuse systems, as well as exploring problem specifications deeply and foundationally to profit from applying tools and methods across disciplines.



Software Engineer

2026/01–present

Salzgitter Digital Solutions GmbH

- software architecture, implementation, and quality assurance of products and pipelines for AI use cases (SE4AI)
- design of workflows, guidelines, and standards for our engineering team
- mentoring and team education

Research Assistant

2020/03–2025/12

Ulm University (2020–24), Paderborn University (2024), TU Braunschweig (2025)

- 30 scientific publications (6 as first-author including 4 top-ranked conferences) in the fields of software engineering, programming languages, computer graphics
- developed several small to large-scale and peer-reviewed research prototypes
- team lead in several (international) research efforts
- supervised multiple bachelor's and master's theses within my research agenda
- teaching assistant for compiler construction, software product lines, software engineering projects, seminars

Student Research Assistant

2019/07–2020/01

Inst. of Software Engineering and Autom. Inf., TU Braunschweig

Designed, implemented, and published [8] a meta-algorithm for reducing a particular SMT problem to SAT in the context of configuring university courses.

Student Research Assistant

2016/11–2020/01

Computer Graphics Lab, TU Braunschweig

Responsible for maintaining and extending C++/OpenGL real-time rendering engines for research on virtual reality, panorama imagery, and animation. I presented the lab's research to pupils once a year.

Other Positions

2014/06–2015/10 *Part-time Junior Software Developer*
Richter Datensysteme GmbH

2015/11–2017/02 *Tutor for Lectures on Algorithms and Data Structures*
Algorithms Department, TU Braunschweig

Awards

- 2024/10 [Distinguished Artifact](#) at OOPSLA'24 *as first author*
- 2024/09 [Best Research Paper](#) at SPLC'24 *as co-author*
- 2024/07 [Best Demonstrations Paper](#) at FSE'24 *as first author*
- 2022/09 [Best Demonstrations Paper](#) at SPLC'21 *as co-author*
- 2021/08 [Best Artifact](#) at ESEC/FSE'21 *as first author*
- 2016/07 Winner of the Software Engineering Student Competition at TU Braunschweig *As a team, we implemented a 3D game engine in Java including an editor and a small game. We presented our product next to all other teams at the uni's exhibition to a jury consisting of professors, company delegates, and student representatives.*

Education

PhD in Computer Science
2020–2025, Ulm University,
summa cum laude

Analyzing Edits to Static Variability
Master in Computer Science
2017–2020, TU Braunschweig
*Semi-Automated Inference of Feature
Traceability During Software
Development*

Bachelor in Computer Science
2014–2017, TU Braunschweig
*Evaluation of Skinning Techniques for
Skeletal Animation in MonSteR*

Communication

- excellent scientific writing
- high-quality documentation
- experienced speaker
- experienced supervisor and teacher

Languages & Tools

Expert: Agda, Python, Java, C/C++, C#, Git, LaTeX
Experienced: Haskell, CI/CD, Linux, real-time rendering, game engine development
Knowledgeable: Kotlin, JS, ANTLR, Nix/NixOS, ...

Math & Formal Methods

Expert: theorem proving, formal semantics, specify and implement PLs, DSLs, compilers, interpreters; software variability
Experienced: type theory, category theory, algebra, logic(s), constraint solving

Languages

German (native), English (fluent)

Portfolio

Vatras Agda     12 ★ formal framework for modeling, studying, and comparing languages for static variability in terms of proven to be correct compilers	DiffDetective Java      9 ★ variability-aware source code differencing and large scale empirical analyses of git repositories	Feature Trace Recording Haskell    5 ★ concept implementation for recording code-to-feature traces from source code edits
Polypropylene C++  19 ★ embedded domain-specific language and dynamic type-system for polymorphoism-aware entity component systems	PaxEngine 3 C++  20 ★ plugin-based 2D/3D game engine based on Polypropylene, OpenGL, SDL, Box2D, and assimp, and including several demos	Centers of Rotation Skinning C++/GLSL  59 ★ shaders and library to compute centers of rotation for a specific 3D skeletal animation method

A full list of my projects can be found on my website pmbittner.github.io/projects/.

Highlighted Publications

- [1] Analyzing Edits to Static Variability
P. M. Bittner, DOI: [10.18725/OPARU-58798](https://doi.org/10.18725/OPARU-58798)
PhD Thesis, 2025
- [2] Decades of GNU Patch and Git Cherry-Pick: Can We Do Better?
A. Schultheiß, A. Boll, P. M. Bittner, S. Greiner, T. Thüm, and T. Kehrer
ICSE, 2026
- [3] On the Expressive Power of Languages for Static Variability
P. M. Bittner, A. Schultheiß, B. Moosherr, J. M. Young, L. Teixeira, E. Walkingshaw, P. Ataei, and T. Thüm
PACMPL no. Object-Oriented Programming, Systems, Languages & Applications (OOPSLA), Oct. 2024, DOI: [10.1145/3689747](https://doi.org/10.1145/3689747),   , **Distinguished Artifact**
- [4] Variability-Aware Differencing with DiffDetective
P. M. Bittner, A. Schultheiß, B. Moosherr, T. Kehrer, and T. Thüm
FSE Companion, Jul. 2024, DOI: [10.1145/3663529.3663813](https://doi.org/10.1145/3663529.3663813),  , **Best Demonstrations Paper**
- [5] Views on Edits to Variational Software
P. M. Bittner, A. Schultheiß, S. Greiner, B. Moosherr, S. Krieter, C. Tinnes, T. Kehrer, and T. Thüm
SPLC, Aug. 2023, DOI: [10.1145/3579027.3608985](https://doi.org/10.1145/3579027.3608985),  
- [6] Classifying Edits to Variability in Source Code
P. M. Bittner, C. Tinnes, A. Schultheiß, S. Viegner, T. Kehrer, and T. Thüm
ESEC/FSE, Nov. 2022, DOI: [10.1145/3540250.3549108](https://doi.org/10.1145/3540250.3549108),  
- [7] Feature Trace Recording
P. M. Bittner, A. Schultheiß, T. Thüm, T. Kehrer, J. M. Young, and L. Linsbauer
ESEC/FSE, Aug. 2021, DOI: [10.1145/3468264.3468531](https://doi.org/10.1145/3468264.3468531),  , **Best Artifact**
- [8] SAT Encodings of the At-Most-k Constraint – A Case Study on Configuring University Courses
P. M. Bittner, T. Thüm, and I. Schaefer
SEFM, Sep. 2019, DOI: [10.1007/978-3-030-30446-1_7](https://doi.org/10.1007/978-3-030-30446-1_7)

A full list of my publications can be found on my website pmbittner.github.io/publications/.